

Answer Key — Physics Grade 10

Sub-Strand 2.1: Properties of Waves • Lesson 1: Anchoring Phenomenon Launch — "The Matatu Horn and the Fading Radio"

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	C	D	A	A	A	A

USE AFTER: PREDICT PHASE — Step 2: Live demonstration with the buzzer on a string

Q1. In the class demonstration, a buzzer playing a steady tone is swung on a string towards the class and then away from the class. Which statement is an OBSERVATION (something you noticed) and not an explanation?

- A. The pitch changed because the person swinging the string pressed the buzzer differently.
- B. The pitch changed because the buzzer's battery was getting weaker.
- C. The pitch sounded higher as the buzzer came towards us and lower as it moved away.**
- D. The pitch changed because the string made the buzzer vibrate faster.

Correct answer: C. An observation describes what was heard or seen. The other choices give reasons (inferences) that nobody has tested yet.

USE AFTER: PREDICT PHASE — Step 1: Hook narration and PREDICT card prompt 1 ("What did you NOTICE?")

Q2. A student travelling by bus from Nakuru to Nairobi enters the long Kabete underpass while the radio is tuned to Classic 105 (FM, 105.2 MHz). What did the student notice about the radio signals in the underpass?

- A. The AM signal cut out, but the FM signal stayed audible.
- B. Both the FM and AM signals cut out completely.
- C. The FM signal cut out almost completely, but the AM signal stayed audible.**
- D. The FM signal became louder and the AM signal stayed the same.

Correct answer: C. The phenomenon states that FM cut out almost completely in the tunnel while AM on the same receiver remained audible.

USE AFTER: OBSERVE PHASE — Station A: PhET Sound Simulation

Q3. At Station A, students use the PhET 'Sound' simulation. They move a listener while the source stays still, then move the source while the listener stays still. Which two things do they record in their observation table for each set-up?

- A. The wavelength of the FM signal and the width of the tunnel.
- B. The name of the wave and the colour of the wave in the simulation.
- C. The temperature of the room and the loudness of the speaker.
- D. The frequency setting and whether the pitch is heard as higher, lower or the same.**

Correct answer: D. The observation table links the frequency setting to the perceived pitch (higher, lower or same) for each configuration. Students only record what they see and hear.

USE AFTER: OBSERVE PHASE — Station B: Diagram Analysis Card (tunnel width and FM/AM wavelength scales)

Q4. A Diagram Analysis Card shows a cross-section of the Kabete tunnel, drawn to scale. The tunnel is about 8 m wide. The FM wavelength (105.2 MHz) is about 2.9 m and the AM wavelength is about 300 m. Which statement correctly compares these measurements? (Only compare the numbers; do not explain anything yet.)

- A. The FM wavelength is smaller than the tunnel width, but the AM wavelength is much larger than the tunnel width.**
- B. The FM wavelength is about 300 m and the AM wavelength is about 2.9 m.
- C. Both wavelengths are larger than the tunnel width.
- D. Both wavelengths are smaller than the tunnel width.

Correct answer: A. 2.9 m is less than 8 m, while 300 m is far greater than 8 m. This is a measurable observation, not an explanation.

USE AFTER: EXPLAIN PHASE — Link back to phenomenon: 'Can our current best explanations fully answer the driving question? What is still missing?'

Q5. After the stations and the group task, a student asks: 'So do we now fully know why FM cuts out in the tunnel but AM does not?' What is the best answer at this point in Lesson 1?

- A. Not yet. We have gathered observations and first ideas, and we will test them in the coming lessons.**
- B. Yes. The PhET simulation has already explained it completely.
- C. No. Physics cannot explain radio signals in tunnels.
- D. Yes. It is because Classic 105 has a weaker transmitter than the AM station.

Correct answer: A. This is the anchor lesson, so the phenomenon is left open. Students have only collected evidence and made first ideas. A 'weaker transmitter' explanation is a technology-only idea that the teacher asks students to re-examine.

USE AFTER: DQB PHASE — Individual sticky-note questions for the Driving Question Board ('The Horn Puzzle' and 'The Tunnel Puzzle' columns)

Q6. Students write questions on sticky notes for the Driving Question Board about the matatu horn and the Kabete tunnel. Which note is the best mechanistic question, one that starts with HOW or WHY and asks about a physical process?

- A. Why does the pitch of a horn change only when the matatu is moving?**
- B. What colour was the matatu?
- C. Which radio station is the best to listen to on the bus?
- D. Who owns the Classic 105 radio station?

Correct answer: A. A good DQB question starts with 'How' or 'Why' and asks about a physical process that can be investigated. The other questions ask about facts or opinions that do not help explain the phenomenon.

USE AFTER: MODEL BUILDING PHASE — Individual 'Initial Wave Model' drawing in notebooks and pair comparison

Q7. Your teacher asks you to draw an 'Initial Wave Model' showing a wave travelling from a matatu horn to your ear and then entering a narrow tunnel. What is the main purpose of this model?

- A. To show your current thinking so that you can revise it as you gather evidence in later lessons.**
- B. To get marks for drawing the correct wave shape.
- C. To give the final scientific explanation of the horn and the radio.
- D. To show how well you can draw a matatu.

Correct answer: A. Students are told that the model does not need to be correct. It records their present ideas so they can compare and revise it as the unit goes on.